

THE V(D)J RECOMBINATION MODEL AND FAST GENERATION PROBABILITIES IN VDJTOOLS

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Appendix M — vjtools model engine

Abstract

The `vdjtools` model engine reimplements the Murugan–Marcou generative model of V(D)J recombination [1, 2] on the AIRR schema [3], storing every model factor as a tidy `polars` table so that the Bayesian-network structure is declared as data rather than baked into a file format. On top of this representation it computes the generation probability (P_{gen}) of a CDR3 — the total probability that the stochastic recombination process produces a given sequence — both for nucleotide queries (an exact scenario sum) and, more delicately, for amino-acid queries, where one must marginalize over all synonymous nucleotide realizations. For the amino-acid case we use the transfer-matrix factorization of [1, 4], but re-derive it from a single structural observation — *that the only coupling across a recombination-block boundary is codon translation, never the insertion Markov chain* — which yields a clean left/right partial-sum split (§M.5) that our native C++ core evaluates faster than OLGA (§M.6) while matching it to machine precision. The same tables and scenario enumeration drive sequence generation (ancestral sampling) and model inference (expectation–maximization), which we summarize in §M.8.

M.1. THE GENERATIVE RECOMBINATION MODEL

A rearranged receptor chain is the output of a stochastic process that (i) selects germline segments, (ii) trims their ends by exonuclease deletion, and (iii) inserts non-templated nucleotides at the junctions. Following [1, 2] we treat a *recombination scenario* r — the full list of hidden events — as a sample from a Bayesian network whose factors are the model parameters. For a VDJ locus (TRB, TRD, IGH) a scenario is

$$r = (V, J, D, \text{del}_V, \text{del}_J, \text{del}_D^{5'}, \text{del}_D^{3'}, \mathbf{m}^{VD}, \mathbf{m}^{DJ}), \quad (1)$$

with probability

$$P(r) = P(V)P(J)P(D|J)P(\text{del}_V|V)P(\text{del}_J|J)P(\text{del}_D^{5'}, \text{del}_D^{3'}|D)P_{\text{ins}}(\mathbf{m}^{VD})P_{\text{ins}}(\mathbf{m}^{DJ}). \quad (2)$$

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A VJ locus (TRA, TRG, IGK, IGL) has no D segment; its scenario drops D , the two D -deletions and one insertion, and the gene factor becomes the joint $P(V)P(J | V)$ with a single VJ insertion. The observed CDR3 nucleotide sequence $\sigma(r)$ is the concatenation, from the conserved Cys 104 to the conserved Phe/Trp 118 inclusive,

$$\sigma(r) = \underbrace{\hat{V}}_{V \text{ germline}} \underbrace{\mathbf{m}^{VD}}_{N1} \underbrace{\hat{D}}_{D \text{ germline}} \underbrace{\mathbf{m}^{DJ}}_{N2} \underbrace{\hat{J}}_{J \text{ germline}}, \quad (3)$$

where $\hat{V}, \hat{D}, \hat{J}$ are the germline segments after the indicated deletions.

INSERTION CHAINS. Each non-templated segment $\mathbf{m} = m_1 \cdots m_\ell$ is drawn from a length distribution and a first-order Markov chain:

$$P_{\text{ins}}(\mathbf{m}) = P_{\text{ins}}(\ell) b(m_1) \prod_{i=2}^{\ell} R(m_i | m_{i-1}), \quad (4)$$

with b the first-nucleotide (stationary) bias and R the 4×4 dinucleotide transition matrix. Following IGoR/OLGA, the VD chain is read $5' \rightarrow 3'$ and the DJ chain $3' \rightarrow 5'$, so b_{DJ} biases the J -proximal insertion nucleotide.

GENERATION PROBABILITY. The quantity of interest is the probability of an *observed* CDR3, obtained by summing over every scenario that could have produced it,

$$P_{\text{gen}}(\sigma) = \sum_{r: \sigma(r)=\sigma} P(r), \quad P_{\text{gen}}(\alpha) = \sum_{\sigma: \tau(\sigma)=\alpha} P_{\text{gen}}(\sigma), \quad (5)$$

where τ is translation: an amino-acid query α additionally marginalizes over all synonymous nucleotide sequences. Both sums are finite but astronomically large if taken literally; §M.3 and §M.5 evaluate them by dynamic programming.

M.2. MODEL PARAMETERS AS DATA

`vdjtools` stores a model as a directory of long-format `polars` parquet tables plus a `manifest.json` that names each event and lists its conditioning parents — the edges of the Bayesian network of Eq. (2). Changing a locus’s conditioning (e.g. $P(D | J)$ versus a richer $P(D | V, J)$) is therefore a one-line manifest edit, not a format change, and a VJ locus is simply the same schema with the D -tables and second insertion absent. Each table holds the distribution of one event, normalized within its conditioning key by a single `polars group_by` — which is exactly the M-step of §M.8. Two conventions matter for the P_{gen} recursions.

PALINDROMIC DELETIONS. A negative deletion count encodes templated palindromic (P-) nucleotides. Each germline cut segment is pre-extended by up to `maxpal` reverse-complement nucleotides; a “deletion” of $-k$ means k P-nucleotides are retained. The stored count `del` is thus *biological* (negative = P-nucleotides), while the dense array index used at run time is `del + maxpal` ≥ 0 .

GERMLINE AS A SINGLE SOURCE OF TRUTH. All V/D/J germline sequences and the CDR3 anchor offsets resolve from `arda` by allele name; `arda`’s anchor convention is byte-identical to OLGA’s (a 0-based Cys 104 / [FW] 118 offset into the full germline), so no coordinate conversion is needed. A model never mixes germline sources: bootstrap models imported from OLGA keep OLGA germline to preserve exact- P_{gen} fidelity, whereas natively-inferred models use `arda`.

THE D-D EXTENSION. The schema carries tandem- D ($n_D \in \{0, 1, 2\}$) rearrangements — a D -count event $P(n_D)$, a second D -gene table $P(D_2 | D_1)$, a joint second- D deletion and a DD insertion — which OLGA/IGoR cannot represent (§M.4). Real 5'-RACE TRD repertoires make this concrete: after collapsing to unique clonotypes, 4.15% of *out-of-frame* and 3.42% of *in-frame* δ -chain clonotypes carry two aligned D segments, the signal on which the $n_D = 2$ event is trained; the corresponding tandem fraction under any single- D model is structurally zero.

M.3. NUCLEOTIDE GENERATION PROBABILITY

For a nucleotide query σ the inner sum of Eq. (5) is a direct enumeration, because the germline segments and both junctions are pinned by σ . Writing $N = |\sigma|$ and letting a V (resp. J) contribute $\ell_V \geq 1$ (resp. $\ell_J \geq 1$) nucleotides — a valid rearrangement is never fully V - or J -deleted, matching OLGA —

$$P_{\text{gen}}(\sigma) = \sum_{V,J} P(V)P(J|V) \sum_{\substack{\ell_V, \ell_J \geq 1 \\ \sigma[0:\ell_V] = \hat{V}, \sigma[N-\ell_J:] = \hat{J}}} P(\text{del}_V)P(\text{del}_J) \Xi(\sigma[\ell_V:N-\ell_J]), \quad (6)$$

where for a VJ locus $\Xi(\mu) = P_{\text{ins}}(\mu)$ is the single-junction insertion weight of Eq. (4), and for a VDJ locus Ξ is itself a sum over the D gene, its two deletions and its placement,

$$\Xi(\mu) = \sum_D P(D|J) \sum_{\text{del}_D^{5'}, \text{del}_D^{3'}} P(\text{del}_D^{5'}, \text{del}_D^{3'}) \sum_{\substack{\text{placements } p \\ \mu[p:p+\ell_D] = \hat{D}}} P_{\text{ins}}(\mu[0:p]) P_{\text{ins}}(\mu[p+\ell_D:]). \quad (7)$$

Common-prefix / common-suffix matching restricts ℓ_V, ℓ_J to the genes that can actually flank σ , so Eq. (6) is fast in practice. This nucleotide sum is the quantity the E-step needs (§M.8); the native port reproduces OLGA's `compute_nt_CDR3_pgen` to machine precision.

M.4. TANDEM-D (D-D) REARRANGEMENTS

A tandem- D locus joins *two* D segments in the junction, $V - D_1 - D_2 - J$, which neither OLGA nor IGoR can represent. We add a D -count event $P(n_D)$, a second gene factor $P(D_2 | D_1)$ (whose zeros carry the genomic-order mask), a joint second- D deletion $P(\text{del}_{D_2}^{5'}, \text{del}_{D_2}^{3'} | D_2)$ and a DD insertion. A two- D scenario has probability

$$P(r_2) = P(n_D=2) P(V)P(J) P(D_1|J)P(D_2|D_1) \prod_{s \in \{V, J, D_1, D_2\}} P(\text{del}_s) P_{\text{ins}}(\mathbf{m}^{VD}) P_{\text{ins}}(\mathbf{m}^{DD}) P_{\text{ins}}(\mathbf{m}^{DJ}), \quad (8)$$

and its CDR3 is $\sigma(r_2) = \hat{V} \mathbf{m}^{VD} \hat{D}_1 \mathbf{m}^{DD} \hat{D}_2 \mathbf{m}^{DJ} \hat{J}$. The middle factor of Eq. (6) becomes a mixture over the D -count,

$$\Xi(\mu) = [P(n_D=0) + P(n_D=1)] \Xi_1(\mu) + P(n_D=2) \Xi_2(\mu), \quad (9)$$

where Ξ_1 is the single- D sum of Eq. (7) and Ξ_2 enumerates two ordered, non-overlapping D placements with VD , DD and DJ insertions between and around them. Two design choices keep the mixture a genuine probability.

PROPOSITION 1 (Disjoint D -count partition). *If every D in a two- D scenario is required to contribute at least one nucleotide ($\ell_{D_1}, \ell_{D_2} \geq 1$), then no CDR3 nucleotide sequence is counted by both Ξ_1 and Ξ_2 , and the $n_D = 0$ (D -less) case is exactly the $n_D = 1$ term with a fully-trimmed D ($\ell_D = 0$). Hence Eq. (9) is a partition of the scenario space and Ξ is normalized whenever the event tables are.*

The first clause removes the identifiability trap — a tandem in which one D is fully deleted is indistinguishable from a single D with a longer insertion — by assigning such realizations to $n_D = 1$ alone; the second means the model needs no separate D -less insertion table (Ξ_1 already sums the $\ell_D = 0$ placement, which is why the loader’s bootstrap $P(n_D) = \delta_1$ reproduces single- D OLGA P_{gen} exactly). The reference implementation of Ξ_2 is a direct enumeration — correct but, like the amino-acid sum of §M.5 before its transfer-matrix rewrite, slow; the planned native port will add one D block to the right partial sum Π_R (Eq. (11)), weighted by $P(n_D = 2)$, so that a tandem query costs one extra block-stitch rather than a nested placement loop. Until that port and the D - D -aware EM land, the not-yet-tandem paths (the native `_core`, amino-acid P_{gen} , generation and inference) refuse a model with $P(n_D = 2) > 0$ rather than silently returning a single- D result; the $n_D = 2$ event is trained by EM on the real TRD clonotypes noted in §M.2.

M.5. AMINO-ACID GENERATION PROBABILITY BY TRANSFER MATRIX

The amino-acid case is harder: the codon marginalization in Eq. (5) reintroduces a sum over the unobserved junction nucleotides. Enumerating scenarios and summing synonymous codons position by position (a codon-constrained dynamic program of width $|\Sigma|$ over the nucleotide state) is correct but costs one $O(N)$ pass per $(D, \text{del}_D^{5'}, \text{del}_D^{3'}, \text{placement})$ skeleton — quadratic in N and prohibitive for long CDR3s. The transfer-matrix method removes the placement and insertion sums.

SETUP. Let $\alpha = \alpha_0 \cdots \alpha_{L-1}$ be the query, $N = 3L$, and σ_i the (summed over) nucleotide at position i . Codon c imposes $\tau(\sigma_{3c}, \sigma_{3c+1}, \sigma_{3c+2}) = \alpha_c$. Because the codon and the insertion Markov chain each look back at most two and one nucleotide respectively, a left-to-right dynamic program with state $x_i = (\sigma_{i-1}, \sigma_{i-2})$ suffices: it can (a) close a codon when $i \equiv 2 \pmod{3}$ and (b) carry the Markov dependency. The width is $|\Sigma| = 25$ (four nucleotides plus a start sentinel, squared). This is the codon-marginalizing DP; the transfer matrix is what lets us reuse its partial results across all scenarios.

THE KEY STRUCTURAL FACT. In the block layout of Eq. (3), $\hat{V} - \mathbf{m}^{VD} - \hat{D} - \mathbf{m}^{DJ} - \hat{J}$, the blocks are coupled only through codon translation, never through an insertion Markov chain.

PROPOSITION 2 (Codon-only coupling). *Across every block boundary in Eq. (3) the recombination weight $P(r)$ factors into a left factor, the indicator(s) of the codon(s) straddling the boundary, and a right factor. No transition $R(\cdot | \cdot)$ spans a boundary.*

Sketch. The VD chain’s leftmost nucleotide m_1^{VD} is drawn from the bias b_{VD} independent of the last V nucleotide, and its rightmost nucleotide carries no transition into $\hat{D}$; the D block is germline (fixed nucleotides, no chain); the DJ chain, read $3' \rightarrow 5'$, draws its J -proximal nucleotide from b_{DJ} independent of the first J nucleotide, and its D -proximal nucleotide carries no transition into $\hat{D}$. Hence

every factor in Eq. (4) lives strictly inside one block. The only inter-block constraints are the codon indicators $\mathbb{1}[\tau(\cdots) = \alpha_c]$ for codons that straddle a boundary. $\square$ $\square$

Proposition 2 licenses computing the left and right halves of the sum *once each* and stitching them, instead of recomputing a full pass per scenario.

LEFT PARTIAL SUMS. Define, for each boundary position p (the position at which $\hat{D}$ starts) and each nucleotide state $x = (\sigma_{p-1}, \sigma_{p-2})$,

$$\Lambda_p(x) = \sum_{\substack{V, \text{del}_V, \ell_{VD} \\ \ell_V + \ell_{VD} = p}} P(V)P(\text{del}_V)P_{\text{ins}}(\ell_{VD}) [VD \text{ Markov}] \prod_{\text{codons} \subset [0, p)} \mathbb{1}[\tau = \alpha], \quad (10)$$

the total weight of all ways the V germline (with $\ell_V \geq 1$) plus the VD insertion fill $[0, p)$ ending in state x , with every codon that lies wholly in $[0, p)$ satisfied. Λ is built by one forward recursion: germline seeds are deposited at $p = \ell_V$, then the VD transfer step — multiply by b_{VD} on the first inserted nucleotide and R_{VD} thereafter, close codons at $p \equiv 2 \pmod{3}$, and add $P_{\text{ins}}(\ell_{VD})$ at each length — extends them, in $O(N|\Sigma|)$ total.

RIGHT PARTIAL SUMS. Symmetrically, for each D gene and boundary p (where $\hat{D}$ ends),

$$\Psi_p^D(y) = \sum_{J, \text{del}_J, \ell_{DJ}} P(D|J)P(J)P(\text{del}_J)P_{\text{ins}}(\ell_{DJ}) [DJ \text{ Markov}] \prod_{\text{codons} \subset [p, N)} \mathbb{1}[\tau = \alpha], \quad (11)$$

with $y = (\sigma_p, \sigma_{p+1})$. Because the DJ chain runs $3' \rightarrow 5'$, Ψ^D is built by a *backward* recursion from the J boundary — one per D gene, the $P(D|J)$ weight folded into the seed. There are at most n_D such passes.

STITCHING. Fixing D , its deletions (hence the fixed germline string $\hat{D}$ of length ℓ_D) and its start p_0 , thread the germline nucleotides of $\hat{D}$ through the DP from Λ_{p_0} to a state distribution at $p_1 = p_0 + \ell_D$ (closing any codons interior to $\hat{D}$, no Markov weight), then contract with $\Psi_{p_1}^D$. By Proposition 2 at most one codon straddles the D/DJ boundary p_1 ; the contraction is a 4×4 (or scalar, when $p_1 \equiv 0$) sum over that codon's free nucleotides. Summing over D , its deletions and p_0 ,

$$P_{\text{gen}}(\alpha) = \sum_D \sum_{\text{del}_D^{5'}, \text{del}_D^{3'}} P(\text{del}_D^{5'}, \text{del}_D^{3'}) \sum_{p_0} \left\langle \mathcal{T}_{\hat{D}} \Lambda_{p_0}, \Psi_{p_1}^D \right\rangle_{p_1}, \quad (12)$$

where $\mathcal{T}_{\hat{D}}$ is the germline-threading operator and $\langle \cdot, \cdot \rangle_{p_1}$ the codon-contracting stitch at p_1 . The insertion-length sums are already inside Λ and Ψ^D , so the outer work is $O(n_D \cdot \#\text{del}_D \cdot N \cdot \ell_D)$ with $\ell_D \ll N$ — versus the enumeration's extra factor of N from re-running the DP per placement. For a VJ locus there is no D and no split: a single codon-marginalizing DP over V -germline – VJ -insertion – J -germline templates suffices, which is already inexpensive.

REMARK 1. This is the $\Pi_L \cdot \Pi_R$ factorization of [1, 4] expressed on the `polars` tables. Where `OLGA` handles the D block with a sliding right-deletion recursion folded into Π_R , we keep Ψ^D purely J/DJ and slide a short explicit germline bridge (Eq. (12)); the two agree to machine precision (§ M.6), and the explicit bridge is what generalizes to the tandem- D event.

M.6. CORRECTNESS AND BENCHMARK

The native transfer matrix is validated against two independent oracles: the transparent amino-acid scenario enumeration of §M.5 (pure Python), and OLGA itself [4]. On the human TRB model it reproduces the published worked example $P_{\text{gen}}(\text{CAWSVAPDRGGYTF} \mid \text{TRBV30}, \text{TRBJ1-2}) = 1.2036 \times 10^{-10}$ and agrees with OLGA to a maximum relative error of 1.2×10^{-15} over 25 marginalized (all V/J/D) and 6 V/J-restricted generated CDR3s — i.e. to machine precision. Table 1 reports timing, measured by `appendix/bench_pgen.py` on the human TRB OLGA default model, single-threaded, over 20 marginalized queries.

Table 1: Amino-acid P_{gen} throughput on human TRB (single thread, marginalized over all V/J/D). TM is the Eq. (12) transfer matrix in the native `_core`; OLGA is `compute_aa_CDR3_pgen`. Lower is better.

Method	ms / sequence	vs OLGA	max. rel. error vs OLGA
OLGA	5.3	1.0×	—
TM (native <code>_core</code>)	0.64	8.3×	1.2×10^{-15}

The pure-Python scenario enumeration of §M.5 is exact and serves as the correctness oracle, but is not a runtime path: unrestricted, it costs on the order of minutes per query (its extra factor of N and its lack of the left/right reuse), which is exactly the cost the transfer matrix removes.

M.7. DEGENERATE, AGNOSTIC AND HAMMING-BALL QUERIES

Three capabilities fall out of the same transfer matrix at essentially no extra cost. First, the codon test at each amino-acid position is a *membership test against an allowed-codon set* (a 64-bit mask), not an equality against one residue; a wildcard mask (any of the 20 amino acids) or any degenerate set is therefore evaluated in a single pass, giving motif/regex P_{gen} without enumerating the matching sequences. Second, conditioning on a V and/or J allele is a usage mask over the gene sums of Eqs. (10), (11); the empty restriction marginalizes over all functional genes — *V-agnostic* / *J-agnostic* P_{gen} .

Combining the two gives a fast one-mismatch query. The total P_{gen} of an amino-acid CDR3 and every sequence within Hamming distance 1 (one substitution) is, by inclusion–exclusion,

$$P_{\text{gen}}^{\leq 1}(\alpha) = \sum_{k=0}^{L-1} P_{\text{gen}}(\alpha_{k \rightarrow *}) - (L-1) P_{\text{gen}}(\alpha), \quad (13)$$

where $\alpha_{k \rightarrow *}$ wildcards position k . OLGA uses the same identity but pays a full (slow) transfer-matrix pass for each of the $L+1$ terms; here each term is one fast pass (Eq. (12)) with a single wildcard mask, so the native `_core` evaluates Eq. (13) at $8.7 \times$ OLGA’s throughput on human TRB (67.9 vs 7.8 ms/query) and $2.5 \times$ on human TRA (25.1 vs 10.1 ms/query), matching it to a relative error of 10^{-15} throughout. A codon-boundary forward/backward sweep would collapse the $L+1$ passes into a single sweep for VJ loci; the D -placement sum couples positions in VDJ loci, so there the $L+1$ -pass identity is retained.

M.8. GENERATION AND INFERENCE ON THE SAME TABLES

GENERATION. Ancestral sampling walks the Bayesian network of Eq. (2) top-down: draw $(V, J[, D])$, the deletions, the insertion lengths, then the insertion nucleotides from the Markov chain, and assemble Eq. (3). Restricting to functional genes and enforcing $\ell_V, \ell_J \geq 1$ keeps every draw scoreable; sampled V/J-usage and CDR3-length distributions match the model marginals.

INFERENCE. Model parameters are estimated from out-of-frame reads by expectation–maximization. The E-step is exactly the scenario sum of §M.3: for each read it computes the posterior weight $P(r \mid \sigma) = P(r)/P_{\text{gen}}(\sigma)$ of every compatible scenario and accumulates soft counts into arrays laid out like the model tables; optionally the enumeration is restricted to the genes `arda` aligns (an *aligned-gene mask*), which prunes the V/D/J sums without changing the fixed point. The M-step renormalizes each event table within its conditioning key — a single `polars group_by` — reproducing the manifest’s Bayesian network. On OLGA-drawn data this is a closed-loop oracle: EM must recover the generating marginals, giving a self-checking target before any real data exists. The E-step inner loop is the second native hot path (the first being P_{gen}); with the aligned-gene mask it makes $V DJ$ inference practical.

M.9. MODEL DIAGNOSTICS: ENTROPY AND MUTUAL INFORMATION

Because the Bayesian network is declared as data (§M.2), every model is directly introspectable: each event table is a (conditional) distribution, so we quantify each part of the rearrangement by its Shannon entropy and each network edge by mutual information, straight from the tables and with no reference to any sequence. For an event X with parents $\text{Pa}(X)$ we report the marginal entropy $H(X)$ (in bits) and, along each declared edge, the information it carries,

$$I(X; \text{Pa}(X)) = H(X) - H(X \mid \text{Pa}(X)), \quad H(X \mid \text{Pa}) = \sum_{\text{pa}} P(\text{pa}) H(X \mid \text{pa}), \quad (14)$$

where parent marginals are obtained by forward propagation over the network. Rendering the manifest as a graph, with nodes annotated by $H(X)$ and edges by I , gives a `bnlearn`-style summary of the whole model at a glance (Fig. 1).

Table 2 compares the per-event entropy across five loci; each row’s most informative locus is in bold. The structure is legible. Gene choice dominates in the $\alpha\beta$ loci ($H_V \approx 5.7$ bits for TRB, $H_J \approx 5.6$ for TRA), whereas TRD spends little on gene choice ($H_V = 2.4$ bits, few δ genes) and the most on non-templated insertion ($H_{VD} = 4.8$ bits) — the TdT signature. The single most entropic event in every $V DJ$ locus is the joint D -deletion (up to 7.8 bits for IGH), and its two ends are far from independent: the within- D coupling $I(\text{del}_D^{5'}; \text{del}_D^{3'} \mid D) = 1.18$ bits for TRB (a genuine conditional mutual information, averaged over D so that the shared choice of D does not inflate it). Two edges are worth naming. First, $I(V; J) = 0$ for every $V DJ$ locus is not a measurement but an *assumption* made visible — OLGA factors $P(V, J) = P(V)P(J)$ — whereas the VJ loci carry $I(V; J) > 0$ through $P(J \mid V)$ (e.g. 0.27 bits for TRA). Second, the D -deletion edge $I(\{\text{del}_D\}; D)$ is large (1.02 bits for TRB), i.e. trimming depends strongly on which D was chosen. The identical routine applies to an EM-inferred model, so the same table compares a model learned from real reads against the OLGA

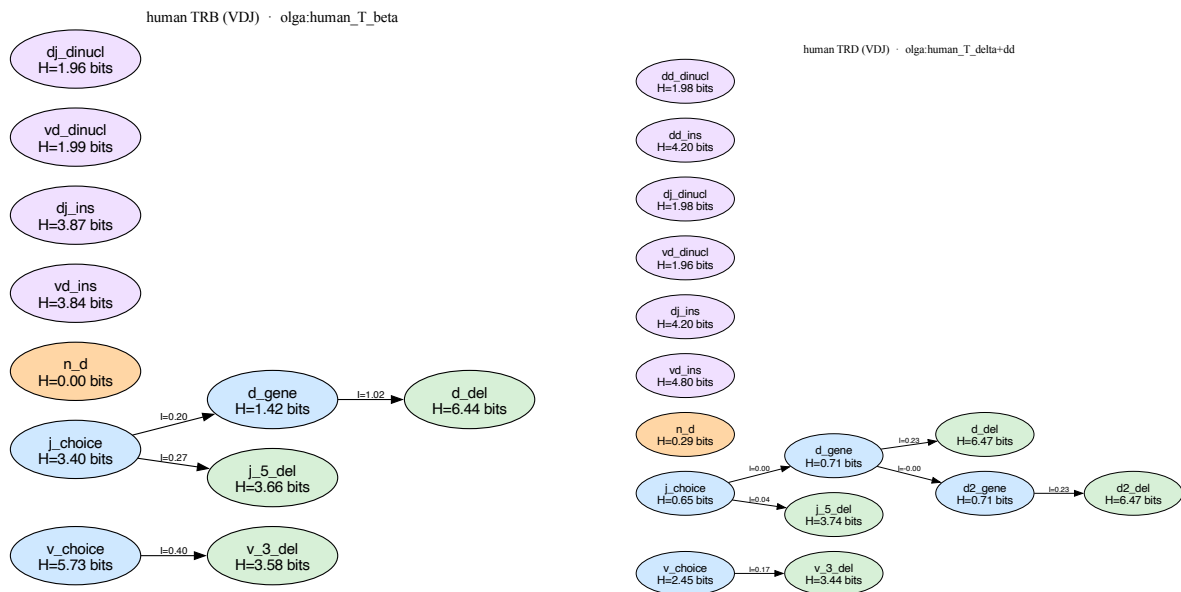


Figure 1: The recombination Bayesian network rendered from the manifest, nodes labelled with marginal entropy H (bits) and edges with mutual information I (bits). *Left*: human TRB (single- D). *Right*: human TRD upgraded to the tandem- D model of § M.4, adding the n_D , D_2 -gene, D_2 -deletion and DD -insertion nodes.

bootstrap parameter-by-parameter. On 6000 real out-of-frame human TRB clonotypes (single- D EM, aligned-gene mask; `bench_em.py`) the fitted model places broader mass on exonuclease deletion and non-templated insertion than the synthetic bootstrap — the D -deletion entropy rises from 6.44 to 7.65 bits and the VD -insertion entropy from 3.84 to 4.50 bits, while the held-out log-likelihood improves — i.e. real repertoires are less tightly trimmed and inserted than the OLGA generative model, and the within- D coupling stays near 1 bit under both.

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Table 2: Marginal entropy $H(X)$ (bits) of each recombination event across five human loci, computed from the model tables. Dashes mark events absent from a locus (VJ loci have no D ; $\alpha\beta$ loci no VJ insertion). Bold marks the most entropic locus per event.

Event	TRB	TRA	TRD	IGH	IGK
V choice	5.73	5.37	2.45	5.71	4.81
J choice	3.40	5.57	0.65	2.17	2.83
D gene J	1.42	—	0.71	4.30	—
V 3' deletion	3.58	3.69	3.44	3.46	3.15
J 5' deletion	3.66	3.89	3.74	4.13	3.65
D deletion (5',3')	6.44	—	6.47	7.78	—
VD insertion length	3.84	—	4.80	4.68	—
DJ insertion length	3.87	—	4.20	4.82	—
VJ insertion length	—	3.84	—	—	3.59
VD dinucleotide	1.99	—	1.96	1.98	—
DJ dinucleotide	1.96	—	1.98	1.99	—
VJ dinucleotide	—	1.97	—	—	1.92

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